

step applied at t equal to zero is therefore given by the integrals

$$i(t) = \int_0^t (e^{-\lambda}) u(t-\lambda) d\lambda + k \int_0^t \int_0^t e^{-\lambda_1 - \lambda_2} u(t-\lambda_1) u(t-\lambda_2) d\lambda_1 d\lambda_2 + 2k \int_0^t \int_0^{\lambda_2} e^{-\lambda_2} u(t-\lambda_1) u(t-\lambda_2) d\lambda_1 d\lambda_2 \quad (37)$$

We have made use here of the symmetry of the h_{qb} function about the line $\lambda_1 = \lambda_2$. The last integration is over half of the positive quadrant of the λ_1, λ_2 plane. The result after integration is

$$i(t) \approx i_1 + i_q = (1 - e^{-t}) - k(1 - 2te^{-t} - e^{-2t}) \quad (38)$$

which is the same result as found in equation 29 by the conventional method.

However, in addition to determining the output of the circuit when a unit step is applied, we have also determined the response functions which apply to *any* input. Thus if the input is $v(t)$, the output will be

$$i(t) \approx \int_0^t e^{-\lambda} v(t-\lambda) d\lambda + k \int_0^t \int_0^t e^{-\lambda_1 - \lambda_2} v(t-\lambda_1) v(t-\lambda_2) d\lambda_1 d\lambda_2 + 2k \int_0^t \int_0^{\lambda_2} e^{-\lambda_2} v(t-\lambda_1) v(t-\lambda_2) d\lambda_1 d\lambda_2 \quad (39)$$

This is a much more general result than that obtained by the conventional method.

This equation can be represented by an equivalent circuit. The first term i_2 is that of an R - C integrator, which is represented by the transfer function $1/(1+s)$. The second term i_{qa} is the same as that of the R - C integrator followed by a multiplier which is discussed in example

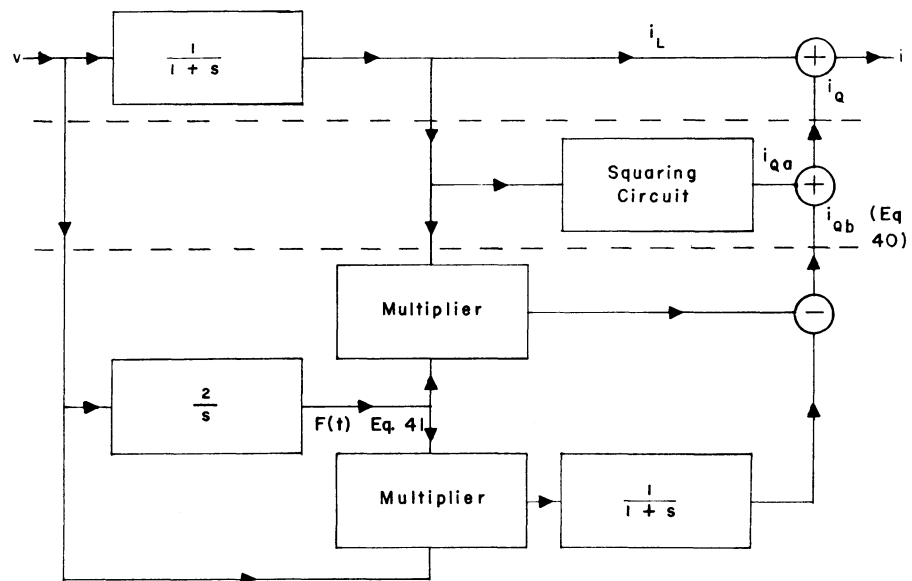


Fig. 6. Equivalent circuit of nonlinear circuit shown in Fig. 4

1. The third term may be rewritten as

$$i_{qb} = \int_0^t e^{-\lambda} [F(t) - F(t-\lambda)] v(t-\lambda) d\lambda = F(t) \int_0^t e^{-\lambda} v(t-\lambda) d\lambda - \int_0^t e^{-\lambda} F(t-\lambda) v(t-\lambda) d\lambda \quad (40)$$

where $F(t)$ is defined by the equation

$$F(t) = 2 \int_0^t v(t-\lambda) d\lambda \quad (41)$$

Equation 40 is represented by the equivalent circuit of Fig. 6, where the overall equivalent circuit of the nonlinear network is also shown.

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On Linear Control Theory

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Synopsis: Optimum control of linear multivariable processes is the concern of this paper. The optimum control problem is studied via the state-variable concept and the dynamic-programing technique. A linear multivariable control system is shown to have optimum performance with respect to a quadratic performance criterion, if both the controller and the estimator are independently designed in an optimum manner.

FOR ABOUT a quarter of a century, the conventional linear control theory has enjoyed a remarkable rate of growth. After years of rapid and active develop-

ment, it appears that the conventional approach of control-system design is reaching its full capability. As a result, control scientists and engineers in this country and in the Soviet Union have launched the exploration of a new philosophy, the search for novel techniques and the development of new theory for control-system design. The research in this direction has been accelerated by the aid of modern high-speed digital computers, and inspired by the inauguration of new branches of applied mathematics, such as dynamic programming. In recent years, optimum

control has become a subject of much interest, a number of papers concerning it having appeared in the literature.

An optimum multivariable control system is generally equipped with an optimum controller and an estimator. The controller is used to generate an optimum control sequence; and the estimator is employed to determine all the state variables from the measurable output variables of the process. The functions of both the controller and the

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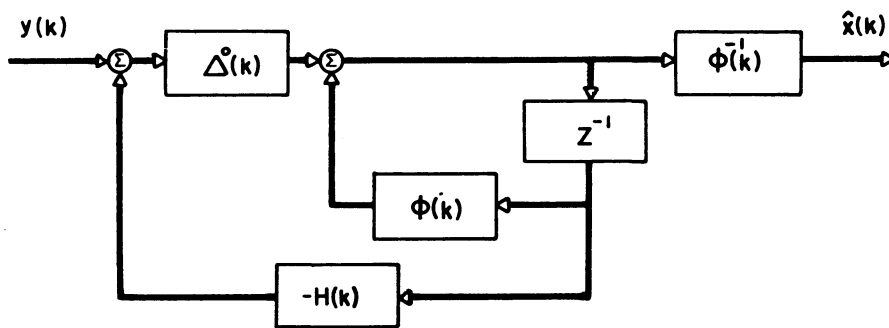


Fig. 1. Optimum estimator when $m = 0$

estimator may be performed by a modern high-speed digital computer. In 1958, Kalman and Koepcke¹ raised the question of "whether the separate optimization of statistical prediction and control-system performance yields a system which is optimal in the *over-all* sense." This paper answers the question in the affirmative for an important class of control problems.

A linear discrete time plant² obeys a vector difference equation

$$\mathbf{x}(k+1) = \Phi(k)\mathbf{x}(k) + \mathbf{A}(k)\mathbf{m}(k) \quad (1)$$

where $\mathbf{x}(k)$ is the state vector, $\mathbf{m}(k)$ is the vector whose components are the control signals, $\Phi(k)$ is the state-transition matrix, and $\mathbf{A}(k)$ determines how the control signal affects the state of the plant.¹⁻⁵

In many problems, it is desirable to choose $\mathbf{m}(k)$ so as to minimize a quadratic performance index of the form

$$I_K = \sum_{k=1}^K \mathbf{x}'(k) \mathbf{Q} \mathbf{x}(k) \quad (2)$$

for any initial $\mathbf{x}(0)$. In equation 2, $\mathbf{Q}$ is a positive definite matrix and the prime denotes the transpose of a vector, or a matrix. The use of dynamic programming⁶ allows the determination of the optimum control law.^{4,7} This law has two important properties.

a. The optimum control signal $\mathbf{m}^\circ(k)$ is a linear function of $\mathbf{x}(k)$, i.e.

$$\mathbf{m}^\circ(k) = \mathbf{B}^\circ(k) \mathbf{x}(k) \quad (3)$$

where $\mathbf{B}^\circ(k)$ is a matrix which can be determined.

b. If the plant is subjected to additive random *white* noise disturbances and the optimum control law, equation 3, is used, the system is still optimal in the sense that $E I_K$ is minimized, where E is the expected value operator.⁷

In order to use equation 3 to generate the control signals, all components of $\mathbf{x}(k)$ must be measurable. In many practical cases, however, not all of the state variables are directly measurable. In more quantitative terms, only certain

signals are directly measurable. These signals will be called the output signals,^{2,3} and a linearly independent set of them will be denoted by $y_1(k), y_2(k), \dots$. For a linear plant, the output signals are linear functions of the state variables, $\mathbf{x}(k)$, i.e.

$$\mathbf{y}(k) = \mathbf{H}(k) \mathbf{x}(k) \quad (4)$$

where $\mathbf{y}(k)$ is the vector whose components are $y_1(k), y_2(k), \dots$.^{2,3} The interesting case occurs when there are fewer output signals than state variables. Then $\mathbf{H}(k)$ is not a square matrix and cannot be inverted.

In order for $\mathbf{m}(k)$ to be *realizable*, it must be a function only of the measurable outputs $\mathbf{y}(k), \mathbf{y}(k-1), \dots$. But the optimum $\mathbf{m}^\circ(k)$ is a function of $\mathbf{x}(k)$. The essential problem then is what is the optimum *realizable* $\mathbf{m}(k)$.

Optimum Estimation

The problem of estimating $\mathbf{x}(k)$ from a knowledge of $\mathbf{y}(k), \mathbf{y}(k-1), \dots$ has been considered in various forms by many authors. The particular formulation of interest here is due to Kalman.^{3,8} The optimum estimate of $\mathbf{x}(k)$ is defined to be the one that minimizes the expected value of the square of the Euclidian norm of $\mathbf{x}(k) - \hat{\mathbf{x}}(k)$, where the estimate is denoted by $\hat{\mathbf{x}}(k)$. When $\mathbf{m}(k) = \mathbf{m}(k-1) = \dots = 0$, the optimum $\hat{\mathbf{x}}(k)$ can be generated by the system of Fig. 1 for a plant subject to a random *white* noise disturbance.⁸ The system consists of a copy of the plant dynamics, to which a correction signal is added. This correction signal is generated by operating on the difference between $\mathbf{H}(k)\hat{\mathbf{x}}(k-1)$ (the "best" estimate of $\mathbf{y}(k)$ given $\mathbf{y}(k-1), \mathbf{y}(k-2), \dots$) and the true value of $\mathbf{y}(k)$ with a matrix $\Delta^\circ(k)$. The optimum $\Delta^\circ(k)$ can be determined by the methods of reference 8.

The optimum estimator has four important properties:³

1. Let $\mathbf{x}(k) - \hat{\mathbf{x}}(k) = \tilde{\mathbf{x}}(k)$. Let $Y(k)$ be the set of all random vectors that can be

expressed as a linear function of $\mathbf{y}(k), \mathbf{y}(k-1), \dots$. Then $\tilde{\mathbf{x}}(k)$ is orthogonal to every vector in $Y(k)$, i.e., if $\mathbf{a}$ is in $Y(k)$ then $E[\mathbf{a}'\tilde{\mathbf{x}}(k)] = 0$.

2. The estimator of Fig. 1 is the optimum linear estimator.

3. If the disturbance is gaussian, it is the optimum estimator.

4. If $\mathbf{m}^\circ(k)$ is given by equation 3, the realizable estimate of $\mathbf{m}^\circ(k)$, $\hat{\mathbf{m}}^\circ(k)$, which minimizes the expected value of the square of the Euclidian norm of $\mathbf{m}^\circ(k) - \hat{\mathbf{m}}^\circ(k)$ is given by

$$\hat{\mathbf{m}}^\circ(k) = \mathbf{B}^\circ(k) \hat{\mathbf{x}}(k) \quad (5)$$

If $\mathbf{m}(k), \mathbf{m}(k-1), \dots$ are not zero, it is easy to show that the optimum estimate is determined by the system of Fig. 2.

When $\mathbf{m}(k), \mathbf{m}(k-1), \dots = 0$, Fig. 1 is the same as Fig. 2 and therefore $\mathbf{x}(k) - \hat{\mathbf{x}}(k)$ is the same $\tilde{\mathbf{x}}(k)$ discussed earlier. When the disturbances and the initial value of $\mathbf{x}(0)$ equals zero and $\mathbf{m}(k), \mathbf{m}(k-1), \dots$ does not equal zero, it is seen by inspection that $\mathbf{x}(k) = \hat{\mathbf{x}}(k)$. Thus, by superposition $\mathbf{x}(k) - \hat{\mathbf{x}}(k) = \tilde{\mathbf{x}}(k)$ and Fig. 2 must be the optimum estimator. Note that $\tilde{\mathbf{x}}(k)$ is independent of $\mathbf{m}(j)$ for all j (Property 5).

The Over-All System

The question now is whether the optimum control law, equation 3, plus the optimum estimator, will yield the optimum over-all system. The answer is yes.

To state this more precisely, when given a linear discrete time plant, let $\mathbf{B}^\circ(k)$ be a sequence of matrices so that the control signal

$$\mathbf{m}^\circ(k) = \mathbf{B}^\circ(k) \mathbf{x}(k) \quad (6)$$

minimizes

$$I_K = \sum_{k=1}^K \mathbf{x}'(k) \mathbf{Q} \mathbf{x}(k) \quad (7)$$

for any arbitrary initial condition.

Given the same linear discrete time plant, whose state variables now are not all measurable and which is subject to a white gaussian disturbance, let $\hat{\mathbf{x}}(k)$ be the vector in $Y(k)$, the set of all linear combinations of the past outputs $y(k), y(k-1), \dots$ which minimizes $E[\|\mathbf{x}(k) - \hat{\mathbf{x}}(k)\|^2]$, where $\|\cdot\|$ denotes the Euclidian norm.

THEOREM

Given a linear discrete time plant, whose state variables are not all measurable and which is subject to an additive white gaussian disturbance, the realizable control signal which minimizes

$$EI_K = E \sum_{k=1}^K \mathbf{x}'(k) \mathbf{Q} \mathbf{x}(k) \quad (8)$$

is given by

$$\hat{\mathbf{m}}^o(k) = \mathbf{B}^o(k) \hat{\mathbf{x}}(k) \quad (9)$$

Proof: Assume momentarily that the disturbance is zero and let

$$J_N = \sum_{j=K-N}^K \mathbf{x}'(j) \mathbf{Q} \mathbf{x}(j) \quad (10)$$

Also let

$$f_N = J_N \quad (11)$$

when $\mathbf{m}(k) = \mathbf{m}^o(k)$

By the principle of optimality, f_N is the absolute minimum of J_N . If there is no disturbance, $J_0 = \mathbf{x}'(K) \mathbf{Q} \mathbf{x}(K)$ is a quadratic form in $\mathbf{x}(K-1)$ and $\mathbf{m}(K-1)$. Thus, J_0 can be written as

$$J_0 = \mathbf{x}'(K-1) \mathbf{P} \mathbf{x}(K-1) + \mathbf{x}'(K-1) \times \mathbf{S} \mathbf{m}(K-1) + \mathbf{m}'(K-1) \mathbf{R} \mathbf{m}(K-1) \quad (12)$$

where $\mathbf{R}$ and $\mathbf{P}$ are symmetric matrices. Since $\mathbf{m}(K-1) = \mathbf{m}^o(K-1)$ minimizes J_0 ,

$$\left. \frac{\partial J_0}{\partial \mathbf{m}_i} \right|_{\mathbf{m}=\mathbf{m}^o} = 0 = \mathbf{x}'(K-1) \mathbf{s}_i + 2\mathbf{m}^o{}' \mathbf{r}_i \quad (13)$$

where $\mathbf{s}_i$ and $\mathbf{r}_i$ are vectors whose components are the components of the i th column of $\mathbf{S}$ and $\mathbf{R}$, respectively. From equation 13

$$\mathbf{x}'(K-1) \mathbf{S} = -2\mathbf{m}^o{}' \mathbf{R} \quad (14)$$

Thus,

$$\begin{aligned} J_0 - f_0 &= \mathbf{m}'(K-1) \mathbf{R} \mathbf{m}(K-1) - \mathbf{m}^o{}'(K-1) \mathbf{R} \mathbf{m}^o(K-1) - 2\mathbf{m}^o{}'(K-1) \mathbf{R} [\mathbf{m}(K-1) - \mathbf{m}^o(K-1)] \\ &= [\mathbf{m}'(K-1) - \mathbf{m}^o{}'(K-1)] \mathbf{R} \times [\mathbf{m}(K-1) - \mathbf{m}^o(K-1)] \quad (15) \end{aligned}$$

Now, if a white gaussian noise disturbance $\mathbf{u}(K-1)$ is added to the plant, J_0 is a quadratic form in $\mathbf{m}(K-1)$, $\mathbf{x}(K-1)$, and $\mathbf{u}(K-1)$.

$$J_0 = \mathbf{x}' \mathbf{P} \mathbf{x} + \mathbf{x}' \mathbf{S} \mathbf{m} + \mathbf{m}' \mathbf{R} \mathbf{m} + \mathbf{u}' \mathbf{F} \mathbf{u} + \mathbf{u}' \mathbf{G} \mathbf{m} + \mathbf{u}' \mathbf{D} \mathbf{x} \quad (16)$$

where the arguments have been dropped temporarily

$$EJ_0 = E[\mathbf{x}' \mathbf{P} \mathbf{x} + \mathbf{x}' \mathbf{S} \mathbf{m} + \mathbf{m}' \mathbf{R} \mathbf{m}] + E[\mathbf{u}' \mathbf{F} \mathbf{u}] \quad (17)$$

and the other terms have dropped out because $\mathbf{u}(K-1)$ is white and, hence, uncorrelated with $\mathbf{x}(K-1)$ and $\mathbf{m}(K-1)$. Then

$$E[J_0 - f_0] = E\{[(\mathbf{m}'(K-1) - \mathbf{m}^o{}'(K-1))] \times \mathbf{R} [\mathbf{m}(K-1) - \mathbf{m}^o(K-1)]\} \quad (18)$$

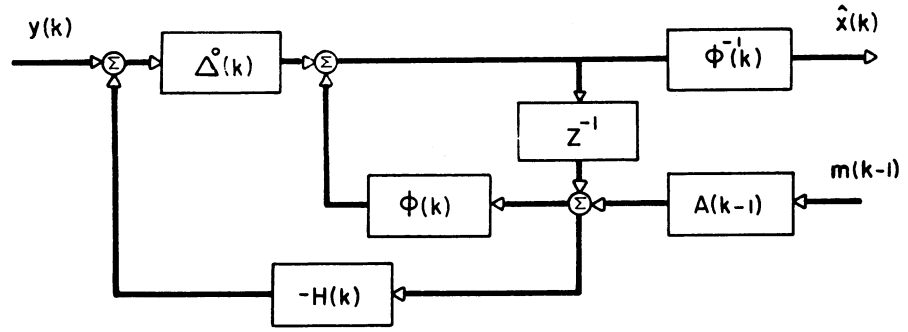


Fig. 2. Optimum estimator when $m \neq 0$

Now $\mathbf{m}(K-1) = \hat{\mathbf{m}}^o(K-1)$ is shown to minimize the foregoing expression and, therefore, J_0 .

First: Since f_0 is the absolute minimum value of J_0 , $J_0 - f_0 > 0$ whenever $\mathbf{m}(K-1) \neq \mathbf{m}^o(K-1)$. Thus, from equation 15 it is obvious that $\mathbf{R}$ is a positive definite matrix.

Second: Let $\mathbf{m}^o(K-1) = \hat{\mathbf{m}}^o(K-1) + \tilde{\mathbf{m}}^o(K-1)$

$$\begin{aligned} E[J_0 - f_0] &= E[(\mathbf{m}' - \hat{\mathbf{m}}^o{}' - \tilde{\mathbf{m}}^o{}') \times \mathbf{R} (\mathbf{m} - \hat{\mathbf{m}}^o - \tilde{\mathbf{m}}^o)] \\ &= E[\tilde{\mathbf{m}}^o{}' \mathbf{R} \tilde{\mathbf{m}}^o + (\mathbf{m}' - \hat{\mathbf{m}}^o{}') \times \mathbf{R} (\mathbf{m} - \hat{\mathbf{m}}^o) - 2\tilde{\mathbf{m}}^o{}' \mathbf{R} (\mathbf{m} - \hat{\mathbf{m}}^o)] \quad (19) \end{aligned}$$

Now, $\tilde{\mathbf{m}}^o(K-1)$ is a linear combination of the $\mathbf{x}(K-1)$ and $\hat{\mathbf{m}}^o(K-1)$ is in $Y(K-1)$. Thus, by property 1

$$E[\tilde{\mathbf{m}}^o{}' \mathbf{R} \hat{\mathbf{m}}^o] = 0 \quad (20)$$

Also for $\mathbf{m}(K-1)$ to be realizable, it must be in $Y(K-1)$. Hence,

$$E[\tilde{\mathbf{m}}^o{}' \mathbf{R} \mathbf{m}] = 0 \quad (21)$$

and

$$E[J_0 - f_0] = E[\tilde{\mathbf{m}}^o{}' \mathbf{R} \tilde{\mathbf{m}}^o + (\mathbf{m}' - \hat{\mathbf{m}}^o{}') \mathbf{R} (\mathbf{m} - \hat{\mathbf{m}}^o)] \quad (22)$$

Since the first term on the right is independent of $\mathbf{m}$, and $\mathbf{R}$ is positive definite, the value of $\mathbf{m}$ which minimizes $E[J_0 - f_0]$ is given by

$$\mathbf{m}(K-1) = \hat{\mathbf{m}}^o(K-1) \quad (23)$$

By the principle of optimality, this is the value of $\mathbf{m}(K-1)$ which minimizes I_K .

To continue

$$\begin{aligned} J_1 &= \mathbf{x}'(K-1) \mathbf{Q} \mathbf{x}(K-1) + J_0 \\ &= \mathbf{x}'(K-1) \mathbf{Q} \mathbf{x}(K-1) + f_0 + [\mathbf{m}'(K-1) - \mathbf{m}^o{}'(K-1)] \mathbf{R} [\mathbf{m}(K-1) - \mathbf{m}^o(K-1)] \quad (24) \end{aligned}$$

In the absence of disturbances, the first two terms on the right are quadratic forms of $\mathbf{x}(K-2)$ and $\mathbf{m}(K-2)$ and,

therefore, in complete analogy with the preceding work.

$$\begin{aligned} E[J_1 - f_1] &= E\{[\mathbf{m}'(K-2) - \mathbf{m}^o{}'(K-2)] \times \mathbf{R}_1 [\mathbf{m}(K-2) - \mathbf{m}^o(K-2)]\} + \\ &E\{[\mathbf{m}'(K-1) - \mathbf{m}^o{}'(K-1)] \times \mathbf{R} [\mathbf{m}(K-1) - \mathbf{m}^o(K-1)]\} \quad (25) \end{aligned}$$

By the principle of optimality, the minimum value of $E[J_1 - f_1]$ is

$$\begin{aligned} \text{Min}_{\mathbf{m}(K-2)} \{ &E[\mathbf{m}'(K-2) - \mathbf{m}^o{}'(K-2)] \times \mathbf{R}_1 [\mathbf{m}(K-2) - \mathbf{m}^o(K-2)] + \\ &E\mathbf{m}^o{}'(K-1) \mathbf{R} \mathbf{m}^o(K-1)\} \quad (26) \end{aligned}$$

Note that by property 5, $\mathbf{m}^o(K-1)$ does not depend upon $\mathbf{m}(K-2)$. Thus, the optimum value of $\mathbf{m}(K-2)$ is the one which minimizes $E[\mathbf{m}'(K-2) - \mathbf{m}^o{}'(K-2)] \mathbf{R}_1 [\mathbf{m}(K-2) - \mathbf{m}^o(K-2)]$. Hence, the optimum $\mathbf{m}(K-2)$ is given by

$$\mathbf{m}(K-2) = \hat{\mathbf{m}}^o(K-2) \quad (27)$$

Clearly, there is now proof by induction that the optimum realizable $\mathbf{m}(k)$ is given by

$$\mathbf{m}(k) = \hat{\mathbf{m}}^o(k) \quad (28)$$

This completes the proof.

REMARKS

1. The theorem allows the design of a control system to be broken into two independent design problems: (1) the determination of the matrix $\mathbf{B}^o(k)$, which is a problem that is solved in reference 1, and (2) the determination of a system to generate $\hat{\mathbf{x}}(k)$. Such a system is shown in Fig. 2 where only the matrix $\Delta^o(k)$ is unknown. The problem of determining $\Delta^o(k)$ is solved in reference 8. Thus, a complete design procedure is available.

2. The theorem could be extended to include performance indices of the form

$$I_K = \sum_{k=1}^K [\mathbf{x}'(k) \mathbf{Q}(k) \mathbf{x}(k) + \mathbf{m}'(k) \mathbf{P}(k) \mathbf{m}(k)] \quad (29)$$

3. By property 2, $\hat{\mathbf{m}}^o(k)$ is the optimum linear-control signal if the disturbance is nongaussian.

4. Nonwhite inputs and disturbances can be handled by considering them to be the outputs of a linear dynamic system excited by white noise.^{8,9} The control system is then designed by considering an *augmented plant*, consisting of the original plant plus the fictitious linear dynamic systems, which generate the noise. Non-regulator problems can be handled by a similar technique.

Conclusions

If the controller and the estimator are independently optimized, an optimum control system, with respect to a quadratic performance criterion, results. Since techniques are already in existence for optimizing controller and estimator, in

single and multivariable, stationary and nonstationary plants, a systematic general method of designing linear discrete data control systems, which are subject to random inputs and disturbances, is available.

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Phase-Space Analysis and Design of Linear Discontinuously Damped Feedback Control Systems

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LINEAR SYSTEMS can be designed having very fast responses with essentially deadbeat performance by the inclusion of compensating loops which are switched in and out of the system as required. Fast response is obtained by using an unstable or nearly unstable uncompensated system to provide the desired rise time. Deadbeat performance is obtained by designing the compensation loops so that they provide an overdamped system when the compensation is switched in. Selection of the real roots for the overdamped system is based on the desired location of the eigenvectors in the phase space. A switching computer is required which connects the compensation loops as the state point reaches a hyperplane which is related to the eigenvector in a special way. The computer is readily realized from derivative signals and considerable engineering simplification is permissible in both the compensation design and the switching computer because of the topological nature of the phase portrait in the vicinity of properly chosen eigenvectors. Experimental results verify the theoretical conclusions.

Nomenclature

ζ = damping factor
 E = error
 θ_R = input quantity
 θ_c = output quantity
 N, n = integers
 A, a = coefficients
 r = root of characteristic equation
 τ = pole of open-loop function
 s = Laplace operator
 P_s = location of the state point in the phase space at the switching instant
 $G(s)$ = open-loop transfer function in the forward path
 $H(s)$ = open-loop transfer function in a feedback path
 K = forward gain of open-loop system

Theoretical Background

It is clear from linear theory that a feedback control system with high gain has fast response and is accurate in both static and steady states. Unfortunately, high gain is usually accompanied by light damping with consequent oscillatory transient response. Various investigators have proposed schemes which essentially vary the damping as some function of the error to obtain small ζ when the error is

large and vice versa. Perhaps the simplest and most obvious approach is that of varying the gain of the main amplifier as suggested by Blumenthal and Beck.¹ Another early and attractive proposal is that of Lewis,² which uses a nonlinear signal to produce a smooth variation in ζ throughout the range of operating inputs. Other proposals have suggested a discontinuous change in ζ . Two such investigations have dealt with relay servos^{3,4} where the damping loop was permanently connected but was operative only in the relay dead zone. Still others have been concerned with switching in a feedback loop. Flügge-Lotz⁵ and Taylor propose to alter both position and velocity feedback in step fashion according to a predetermined schedule which is implemented by a fairly complex computer. Meiksin⁶ proposes switching in main feedback path from positive feedback for fast rise time to negative feedback for good damping. For second-order systems switching along an eigenvector is easily instrumented, but gain design suitable for overdamped dynamic response may not be compatible with static accuracy requirements. For third-order systems eigenvector (for a mathematical definition see reference 7) switching is recommended. (In terms of phase-space trajectories it may be said: A linear differential equation has only one singular point; for each real root of the

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